1 Introduction
These Guidelines contain information and suggestions concerning technical assessments which may be of use in conjunction with the planning of renewal surveys of oil tankers. As indicated in 5.1.5, these Guidelines are a recommended tool which may be invoked at the discretion of an Administration, when considered necessary and appropriate, in

---CHUNK---

SOURCE: MSC.197(80).pdf
As indicated in 5.1.5, these Guidelines are a recommended tool which may be invoked at the discretion of an Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.
2 Purpose and principles
2.1 Purpose
The technical assessments described in these Guidelines should assist in identifying critical structural areas, nominating suspect areas and in focusing attention on structural elements or areas of structural elements which may be particularly susceptible to, or evidence a history of, wastage or damage. This information may be useful in nominating locations, areas and tanks for thickness measurement, close-up survey and tank testing.
2.2 Minimum requirements
These Guidelines may not be used to reduce the requirements of annexes I, 2 and 3 for close-up survey, thickness measurement and tank testing, respectively, which are, in all cases, to be complied with as a minimum.
2.3 Timing
As with other aspects of survey planning, the technical assessments described in these Guidelines should be completed out by the owner or operator in co-operation with the

---CHUNK---

SOURCE: MSC.197(80).pdf
cases, to be complied with as a minimum.
2.3 Timing
As with other aspects of survey planning, the technical assessments described in these Guidelines should be completed out by the owner or operator in co-operation with the Administration well in advance of the commencement of the renewal survey, i.e., prior to commencing the survey and normally at least 12 to 15 months before the survey's completion due date.
2.4 Aspects to be considered
Technical assessments, which may include quantitative or qualitative evaluation of relative risks of possible deterioration, of the following aspects of a particular ship may be used as a basis for the nomination of tanks and areas for survey of:
.1 design features such as stress levels on various structural elements, design details and extent of use of high-tensile steel (HTS);
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 97-
.2 former history with respect to corrosion, cracking, buckling, indents and

---CHUNK---

SOURCE: MSC.197(80).pdf
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED

.2 former history with respect to corrosion, cracking, buckling, indents and repairs for the particular ship as well as similar vessels, where available; and
.3 information with respect to types of cargo carried, use of different tanks for cargolballast, protection of tanks and condition of coating, if any.
Technical assessments of the relative risks of susceptibility to damage or deterioration of various structural elements and areas should be judged and decided on the basis of recognized principles and practices, such as may be found in references I and 2.
3 Technical assessment
3.1 General
3.1.1 There are three basic types of possible failure which may be the subject of a technical assessment in connection with the planning of surveys: corrosion, cracks and buckling. Contact damages are not normally covered by the survey plan since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by surveyors.

---CHUNK---

SOURCE: MSC.197(80).pdf
buckling. Contact damages are not normally covered by the survey plan since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by surveyors.
3.1.2 Technical assessments performed in conjunction with the survey planning process should, in principle, be as shown schematically in figure 1 which depicts how technical assessments can be carried out in conjunction with the survey planning process. The approach is basically an evaluation of the risk, based on the knowledge and experience related to design and corrosion.
3.1.3 The design should be considered with respect to structural details which may be susceptible to buckling or cracking as a result of vibration, high stress levels or fatigue.
3.1.4 Corrosion is related to the ageing process and is closely connected with the quality of corrosion protection at newbuilding and subsequent maintenance during the service life. Corrosion may also lead to cracking and/or buckling.
3.2 Methods
3.2.1 Design details
3.2.1.1 Damage experience related to the ship in question and similar ships, where available, are the main source of information to be used in the process of planning. In

---CHUNK---

SOURCE: MSC.197(80).pdf
3.2 Methods
3.2.1 Design details
3.2.1.1 Damage experience related to the ship in question and similar ships, where available, are the main source of information to be used in the process of planning. In addition, a selection of structural details from the design drawings should be included. Typical damage experience to be considered will consist of:
.1.1 number, extent, location and frequency of cracks; and
.1.2 location of buckles.
3.2.1.2 This information may be found in the survey reports and/or the owner's files, including the results of the owner's own inspections. The defects should be analysed, noted and marked on sketches.

(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 98-
3.2.1.3 In addition, general experience should be utilized. For example, reference should be made to reference I, which contains a catalogue of typical damages and proposed repair methods for various tanker structural details.
3.2.1.4 Such figures should be used together with a review of the main drawings, in

---CHUNK---

SOURCE: MSC.197(80).pdf
should be made to reference I, which contains a catalogue of typical damages and proposed repair methods for various tanker structural details.
3.2.1.4 Such figures should be used together with a review of the main drawings, in order to compare with the actual structure and search for similar details which may be susceptible to damage. An example is shown in figure 2. In particular, chapter 3 of reference 1 deals with various aspects specific to double hull tankers , such as stress concentration locations , misalignment during construction, corrosion trends, fatigue considerations and areas requiring special attention, which should be considered in working out the survey planning.
3.2.1.5 The review of the main structural drawings , in addition to using the above-mentioned figures , should include checking for typical design details where cracking has been experienced. The factors contributing to damage should be carefully considered.
3.2.1.6 The use of HTS is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage

---CHUNK---

SOURCE: MSC.197(80).pdf
considered.
3.2.1.6 The use of HTS is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage when HTS, and its higher associated stresses, are utilized. There is extensive and, in general, good experience, with the use of HTS for longitudinal material in deck and bottom structures. Experience in other locations, where the dynamic stresses may be higher, is less favourable, e.g. side structures .
3.2.1. 7 In this respect, stress calculations of typical and important components and details , in accordance with relevant methods , may prove useful and should be considered.
3.2.1.8 The selected areas of the structure identified during this process should be recorded and marked on the structural drawings to be included in the survey programme .
3.2.2 Corrosion
3.2.2. I In order to evaluate relative corroSIOn risks, the following information IS generally to be considered:
.1.1 usage of tanks and spaces;
· 1.2 condition of coatings;
· 1.3 condition of anodes;
· I .4 cleaning procedures;
· 1.5 previous corrosion damage;
· 1.6 ballast use and time for cargo tanks;

---CHUNK---

SOURCE: MSC.197(80).pdf
generally to be considered:
.1.1 usage of tanks and spaces;
· 1.2 condition of coatings;
· 1.3 condition of anodes;
· I .4 cleaning procedures;
· 1.5 previous corrosion damage;
· 1.6 ballast use and time for cargo tanks;
.1. 7 corrosion risk scheme (see reference 2, table 2.1); and
.1. 8 location of heated tanks.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 99-
3.2.2.2 Reference 2 gives definitive examples which can be used for judging and describing coating condition, using typical pictures of conditions.
3.2.2.3 The evaluation of corrosion risks should be based on information in reference 2, together with the age of the ship and relevant information on the anticipated condition as derived from the information collected in order to prepare the survey programme .
3.2.2.4 The various tanks and spaces should be listed with the corrosion risks nominated accordingly. Special attention should be given to the areas where the double